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SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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SDC HACKATHON 2026

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1. Team & Project Overview

Team Name	PostDepresSQL
Selected Domain	Domain 1
Selected SDG Track (1–17)	SDG 4
Project / Solution Name	SkillShift AI
Name (Leader)	SARVEZ ELANGOVAN
Email & Phone (Leader)	sarvezelangovan69@gmail.com - +60167229286
University Affiliation(s)	Tunku Abdul Rahman University of Management and Technology (TARUMT)
Number of Team Members	4

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

Artificial intelligence is rapidly reshaping job responsibilities and the technical skills expected from graduates and existing workers. However, Malaysian university students and fresh graduates often struggle to identify which competencies they already possess, which emerging skills they lack, and which learning pathways are most relevant to their career aspirations.

The Aiversecity Domain 1 challenge brief highlights this disparity, citing the Hays Asia Salary Guide 2025: 56% of surveyed professionals were already using AI at work, while only 26% had received formal AI training. This illustrates the need for accessible and practical upskilling guidance.

Conventional career guidance often relies on static occupational information or generic recommendations that do not adequately explain skill requirements, the source of workforce evidence, or uncertainty surrounding AI adoption. Consequently, learners may invest time in unsuitable courses or pursue career pathways without understanding their readiness.

This challenge is both educational and economic. Unequal access to relevant skills guidance can undermine lifelong learning, career adaptability, and preparation for future employment.

SkillShift AI addresses this problem by helping learners understand their existing capabilities, explore suitable career options, identify skill gaps, and make informed reskilling decisions using explainable analysis and attributable learning resources.

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

SkillShift AI is a web-based career-readiness and personalized reskilling platform designed primarily for Malaysian university students and fresh graduates.

Users create a profile containing their educational background, interests, and existing skills. A deterministic weighted matching engine analyzes the profile against a curated occupational and skills catalog, generating ranked career-readiness scores, matched competencies, and missing skills.

The platform also presents dated Malaysian workforce observations and their sources where comparable evidence is available. Its scenario simulator lets users examine hypothetical moderate and rapid AI-adoption assumptions rather than presenting uncertain predictions as established facts.

An interactive What-If Simulator allows users to preview how acquiring a missing skill could change their readiness score. This helps learners prioritize relevant competencies before investing in additional training.

SkillShift AI further generates ordered learning roadmaps using approved courses with estimated study times. Optional Qwen-based inference hosted through RunPod can personalize the selection or ordering of approved resources. Backend validation prevents the model from introducing unapproved course identifiers, while an immediate deterministic roadmap remains available without GPU inference.

The application uses React and TypeScript for the frontend, Python FastAPI for backend services, Supabase for authentication and account data, and Replit for public deployment. The solution applies concepts relevant to the hackathon's GenAI, database architecture, and cloud deployment workshops.

The platform primarily supports SDG 4, particularly Target 4.4 concerning relevant skills for employment, with supporting relevance to SDG 8.

SkillShift AI assists human career decisions; it does not make hiring decisions or claim to predict individual employment outcomes.

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

Frontend: React 19, TypeScript, Vite, and the chart/UI libraries included in the application.

Backend: Python 3.12, FastAPI, Pydantic, and the team's deterministic career-matching and scenario-analysis modules.

Database and authentication: Supabase Authentication and PostgreSQL, with Row Level Security for user-owned records.

Deployment: Replit, hosting the compiled React frontend and FastAPI application through a single public HTTPS deployment.

Optional AI: Qwen language model on RunPod for approved-course learning-roadmap personalization.

Data: Curated career, skills, learning-resource, and workforce-evidence catalogs, including attributed Malaysian sources where supported.

System flow:

User → React Interface → FastAPI Backend → Career-Matching Engine → Career Readiness and Skill Gaps → Scenario / What-If Analysis → Approved Learning Roadmap

Additional integrations:

- React / FastAPI ↔ Supabase Auth and PostgreSQL
- FastAPI → RunPod Qwen inference (optional)
- FastAPI → Curated workforce evidence and course catalogs

Sponsor-resource utilization: Replit is used for application deployment. RunPod is used for GPU-based model inference if confirmed by the team's deployment evidence. The standard roadmap functions independently of RunPod.

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

The team adopts a learner-centred design approach, focusing on Malaysian students and fresh graduates who need understandable and actionable guidance when selecting future career pathways.

During problem framing, we identified three user questions:

- *Which skills do I already have?*
- *Which future careers match these skills?*
- *What should I learn next?*

These questions shaped a progressive user journey consisting of profile intake, explainable career matching, hypothetical scenario exploration, skill-gap identification, and personalized learning recommendations.

For feasibility, the team selected a bounded, curated career-and-skills dataset instead of attempting unsupported large-scale job forecasting. The weighted matching algorithm makes readiness results reproducible and explainable, while optional AI personalization is constrained to approved learning resources.

The application separates its React frontend, FastAPI services, Supabase account storage, and optional RunPod inference into maintainable components. Replit provides accessible public deployment, allowing the team to test the application beyond local development.

The current MVP has undergone automated backend testing, with 165 tests passing in the latest reported local run. The next evaluation stage will focus on usability, evidence traceability, learning-roadmap completeness, and reliable interaction with the publicly deployed prototype.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

*The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess **Innovation and Creativity** as well as **Feasibility of build**. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	**Your Response**
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	SDG Target 4.4: Substantially increase the number of youth and adults with relevant skills for employment, decent jobs and entrepreneurship. Supporting connection to SDG 8.
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	The Domain 1 brief cites 56% workplace AI use versus 26% formal AI training in the Hays Asia Salary Guide 2025. These are contextual workforce indicators, not a measured baseline for SkillShift users.
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	Actionable Pathway Completion Rate (%): The percentage of predefined fictional user scenarios in which SkillShift produces an explainable career match, at least one relevant approved learning resource, and a visible SDG connection.
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	Target: at least 90% completion across 10 predefined fictional learner scenarios by the final prototype evaluation. This is a proposed engineering/usability target, not an achieved impact claim.

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

- ☑ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.
- ☑ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.
- ☑ The proposed solution targets a functional, deployed application rather than a theoretical business case study.
- ☑ All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☑ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☑ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	**Type (code/dataset/model)**	**License**
React / FastAPI / Pydantic	Open-source code	MIT — verify installed versions
Vite / TypeScript	Open-source development tools	MIT / Apache-2.0 respectively
DOSM published evidence	Public source data	Cite original source and its applicable terms
Qwen model, if used	Pretrained AI model	Exact model license to be confirmed by Member 3
Microsoft Learn content links	Learning-resource references	Provider terms apply; not relicensed by the team

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☒ No

If yes, please describe the nature of the agreement and which member it concerns:

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.



[24th October 2026]